DS Module Summary

2110403 Introduction to Data Science and Data Engineering (2026/1)

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Prof. Peerapon Vateekul, Ph.D.
Department of Computer Engineering,
Faculty of Engineering, Chulalongkorn University
Peerapon.v@chula.ac.th
www.cp.eng.chula.ac.th/~peerapon/

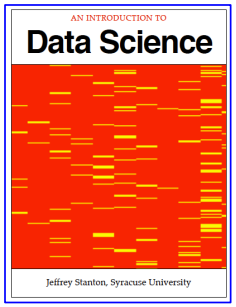
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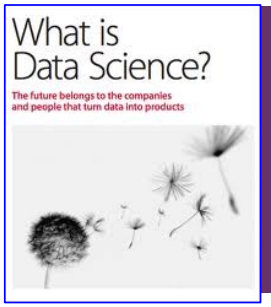
Lots of Topics: DS/DE/Viz

W	#	DoW	Date		Contents	Instructor	Module
10	1	Tue	6-Oct-26	9-12AM	Introduction + Basic Python	Aj.Peerapon	DS1
10	2	Wed	7-Oct-26	9-12AM	TA session	Aj.Peerapon	
				1-4PM	Data Prep	Aj.Peerapon	DS2
11	3	Tue	13-Oct-26	9-12AM	TA session	Aj.Peerapon	
11	4	Wed	14-Oct-26	9-12AM	Traditional ML (1)	Aj.Peerapon	DS3
				1-4PM	Traditional ML (2)	Aj.Peerapon	DS4
12	5	Tue	20-Oct-26	9-12AM	TA session	Aj.Peerapon	
12	6	Wed	21-Oct-26	9-12AM	Traditional ML (3)	Aj.Peerapon	DS5
				1-4PM	Traditional ML (4)	Aj.Peerapon	DS6
13	**7**	**Tue**	**27-Oct-26**	**9-12AM**	**Midterm Practice (On-site, required)**	**Aj.Peerapon**	
13	8	Wed	28-Oct-26	9-12AM	Deep Learning	Aj.Peerapon	DS7
				1-4PM	Advanced topics (Prompt Engineering, RAG, Agentic AI)	Aj.Peerapon	DS8
					Midterm exam on Sun 1 Nov 2026 for 2 rounds (8.30 - 11.30 & 13.00 - 16.00)		
14	8	Tue	3-Nov-26	9-12AM	Intro to Viz + BI + Storytelling	Aj.Veera	VIZ1
14	9	Wed	4-Nov-26	9-12AM	Big data architecture + data storage	Aj.Natawut	DE1
				1-4PM	TA session	Aj.Natawut	
15	10	Tue	10-Nov-26	9-12AM	Web scraping	Aj.Natawut	DE2
15	11	Wed	11-Nov-26	9-12AM	Data ingestion	Aj.Natawut	DE3
				1-4PM	TA session	Aj.Natawut	
16	12	Tue	17-Nov-26	9-12AM	Big data processing (Spark)	Aj.Natawut	DE4
16	13	Wed	18-Nov-26	9-12AM	Intro to Viz + BI + Storytelling	Aj.Veera	VIZ1
				1-4PM	TA session		
17	14	Tue	24-Nov-26	9-12AM	Python visualization	Aj.Veera	VIZ2
17	15	Wed	25-Nov-26	9-12AM	Graph analysis & spatial analysis	Aj.Veera	VIZ3
				1-4PM	**Final Practice (On-site, required)**	**All prof.**	
18	16	Tue	1-Dec-26		**No Class**		
18		Wed	2-Dec-26		**Final Exam on Wed 2 Dec 2026 for 2 rounds (8.30 - 11.30 & 13.00 - 16.00)**		

Attendance	5%
Midterm Exam	30% (Lab Test)
Final Exam	30% (Lab Test)
Assignment	20%
Project	15%

What is Data Science (DS)? (aka. data analytics)



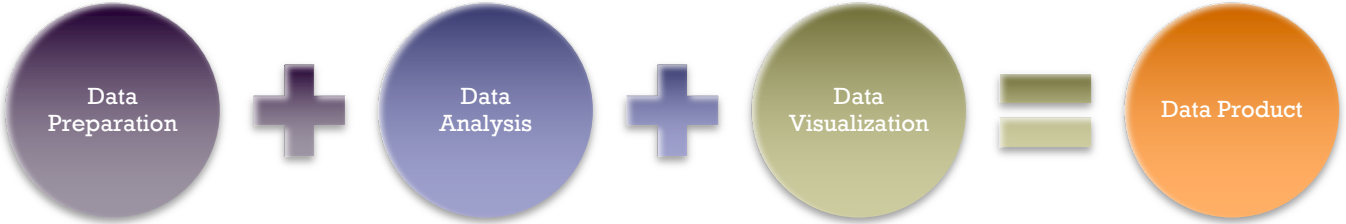


- Data
  - Facts and statistics collected for reference or analysis
- Science
  - A systematic study through observation and experiment
- Data Science
  - The scientific exploration of data to extract meaning or insight,
  - and the construction of software to utilize such insight in a business context.

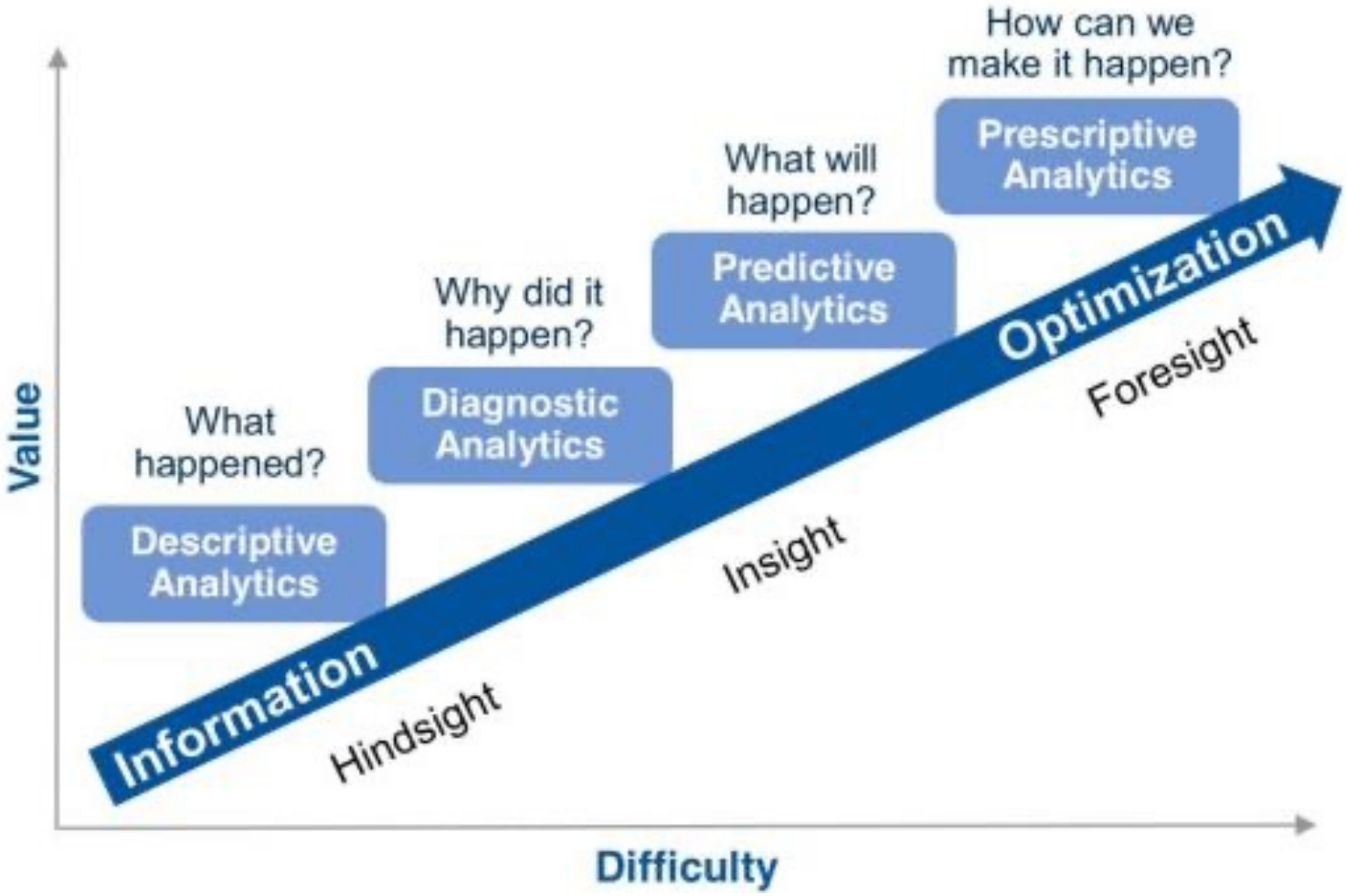
1. Transform data into valuable insights
2. Transform data into data products
3. Transform data into interesting stories

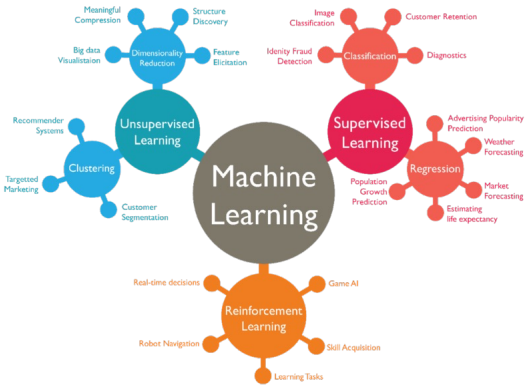


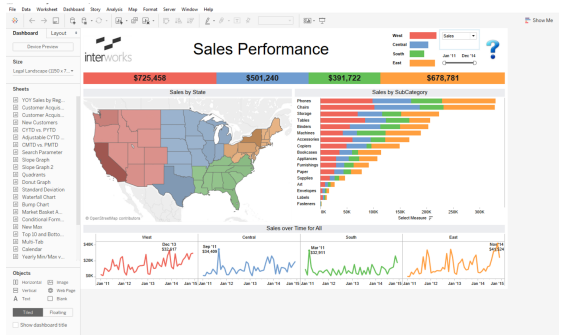
Ta Virot Chiraphadhanakul
Data Scientist, Facebook



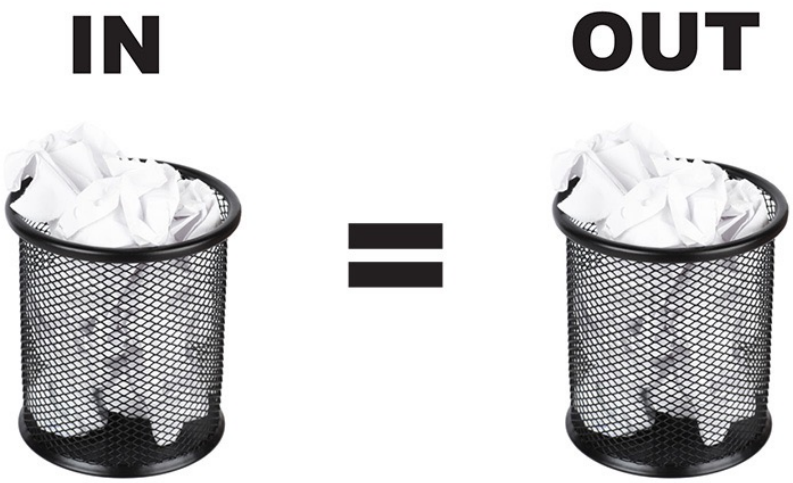
Data Analytics (Data Science)

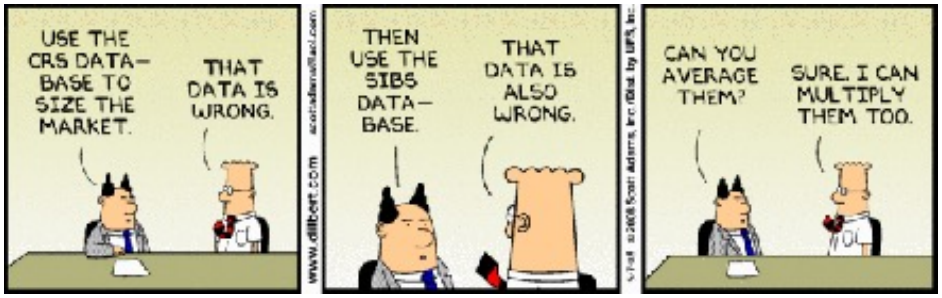


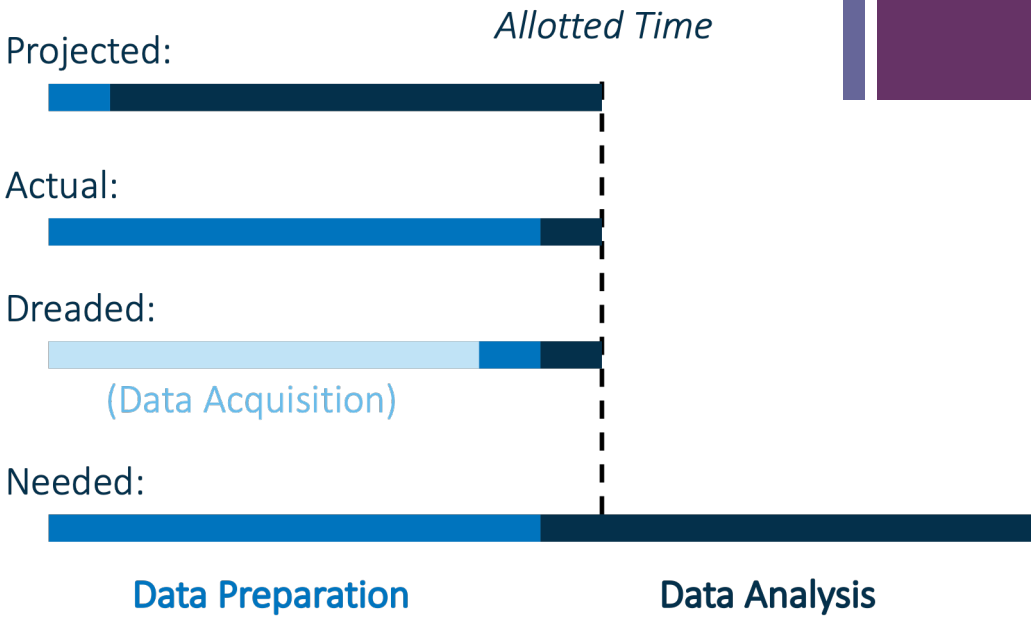




Data is the most important factor. The first thing you should present is not the model, but the data.







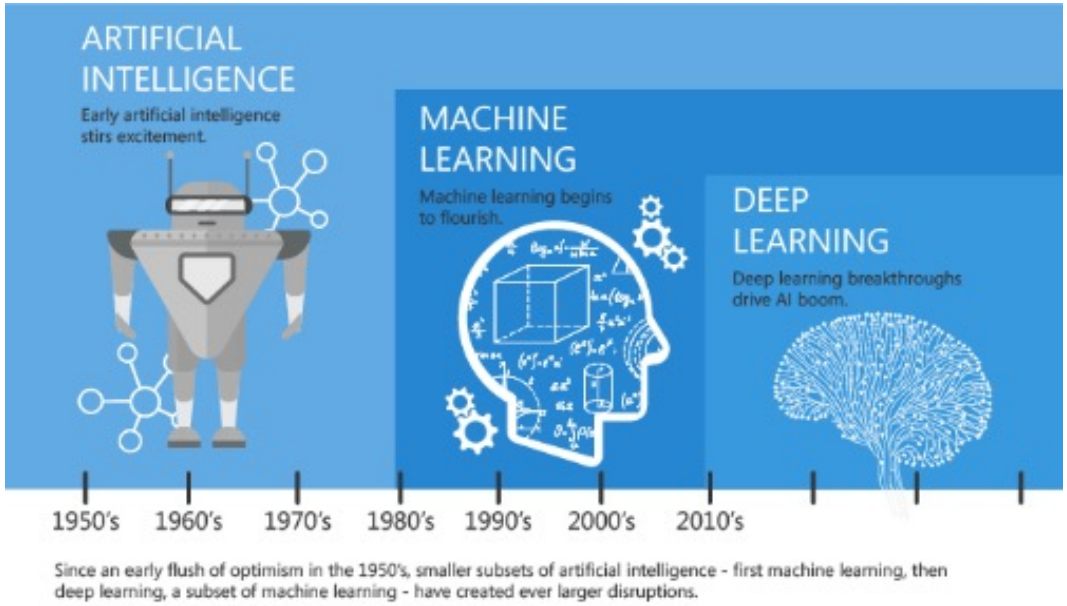


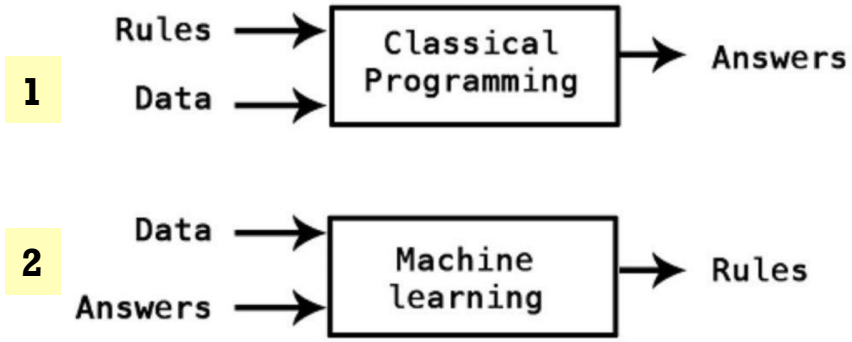
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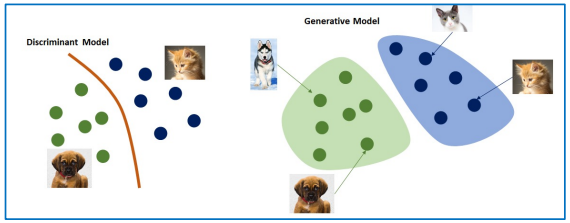
+

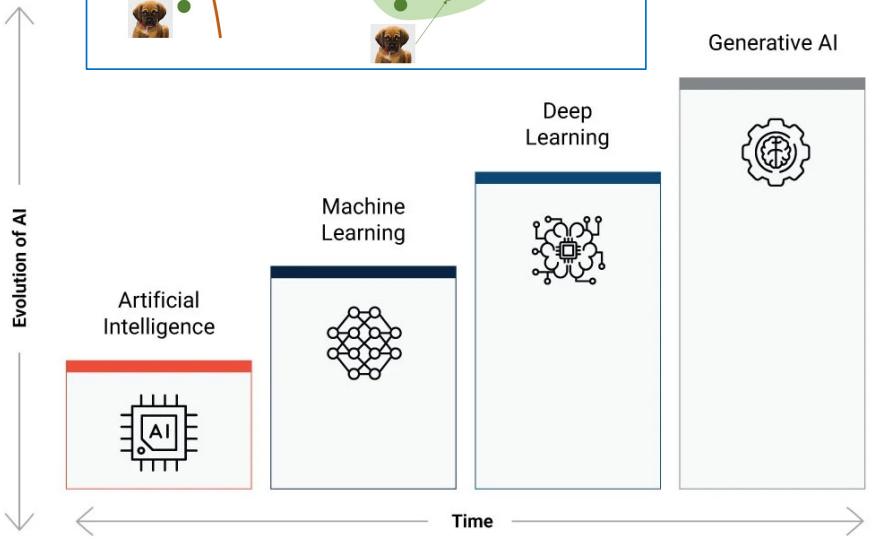
AI = Automation

- 1) Rule-based AI
- 2) Machine Learning (ML)



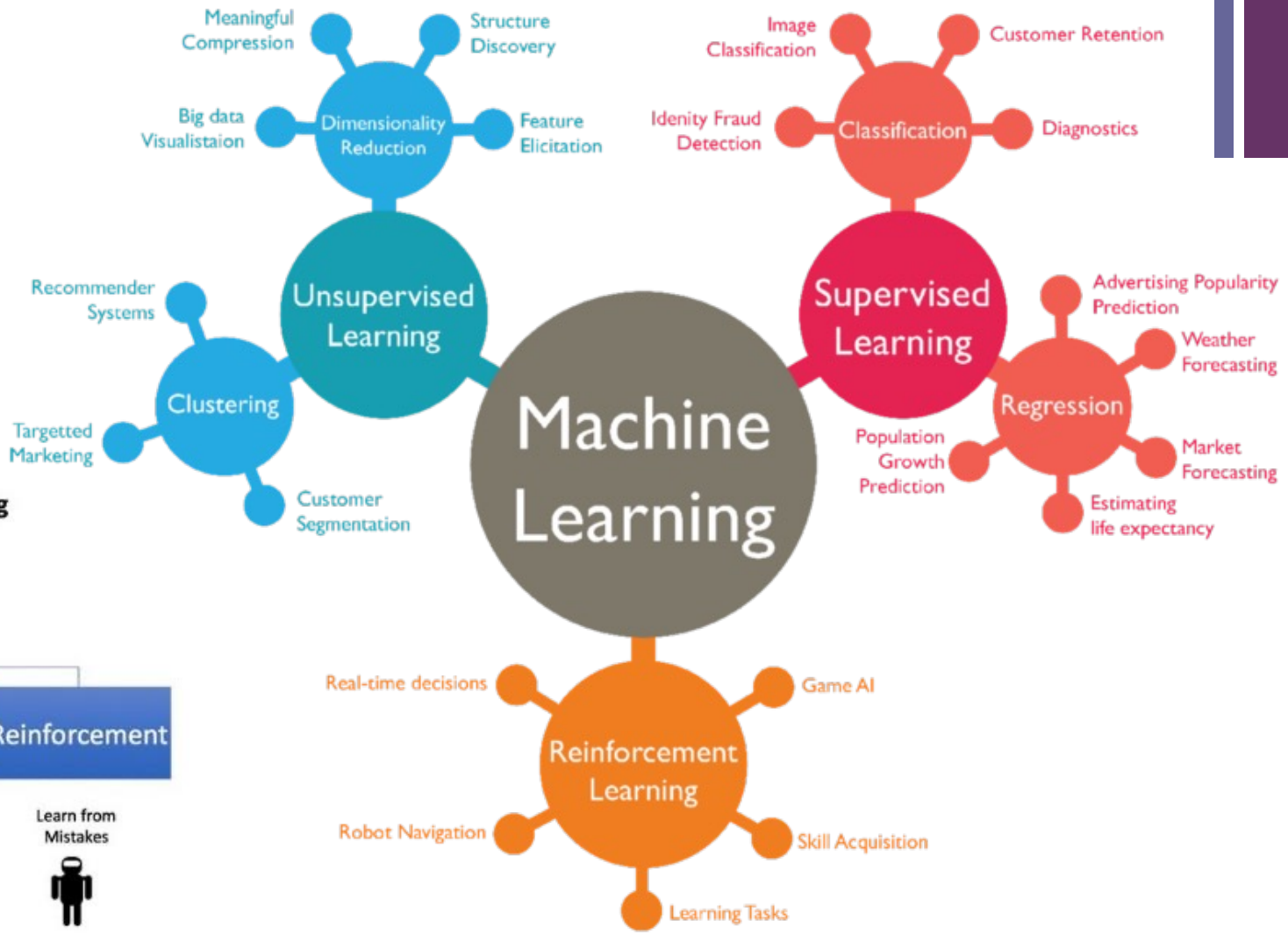


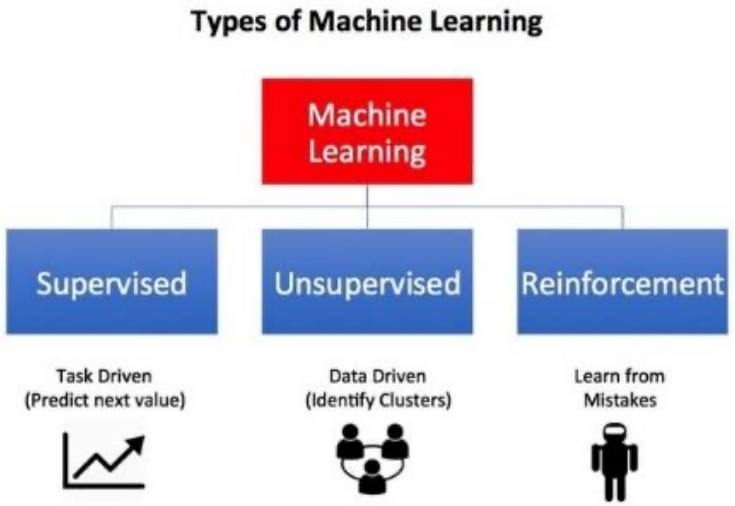




https://mc.ai/machine-learning-basics-artificial-intelligence-machine-learning-and-deep-learning/

Machine Learning (cont.)





Prediction Algorithms

- Decision Tree
- (Logistic) Regression
- Neural Networks (NN)
- kNN
- Support Vector Machine
- Deep Learning

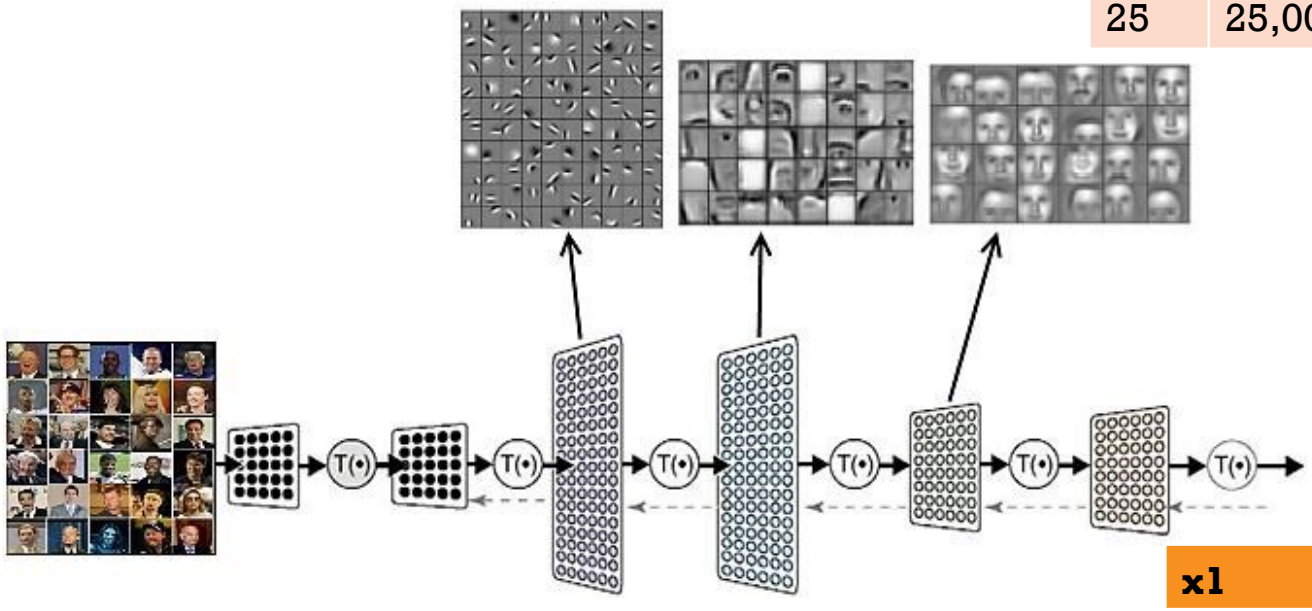


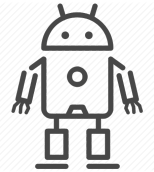
Deep Learning

A deep neural network consists of a hierarchy of layers, whereby each layer transforms the input data into more abstract representations (e.g., edge -> nose -> face). The output layer combines those features to make predictions.



Age	Income	Gender	Province	Corona
25	25,000	Female	Bangkok	Yes





x1	x2	x3	x4	Corona
0.7	0.2	-0.5	-0.1	Yes

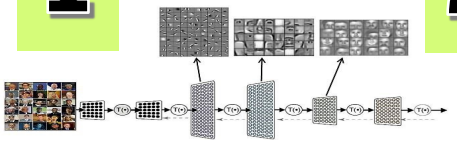
Deep Learning Application

Speech Recognition

Computer Vision

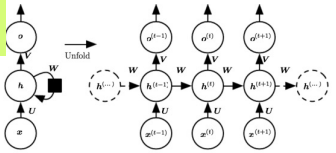
Natural Language Processing

1



CNN

2



RNN

Large + LM = LLM

"Large" is no longer just about parameter count.

Model	Year	Parameters	Key advancement
GPT-1	2018	117M	Transformer language modeling
GPT-2	2019	1.5B	Stronger zero-shot generation
GPT-3	2020	175B	In-context & few-shot learning
GPT-4	2023	ND (~1.7TB)	Multimodality + stronger reasoning
GPT-5	2025	ND (~300B)	Integrated reasoning + tool use
GPT-6 Astra	2026	Not disclosed (ND)	Computer use + end-to-end ag

Modern LLM capability ≠ parameter count alone

Pretraining + Architecture + Post-training + Inference-time Reasoning + Tools

Note: OpenAI has not publicly disclosed parameter counts for GPT-4, GPT-5, or GPT-6.

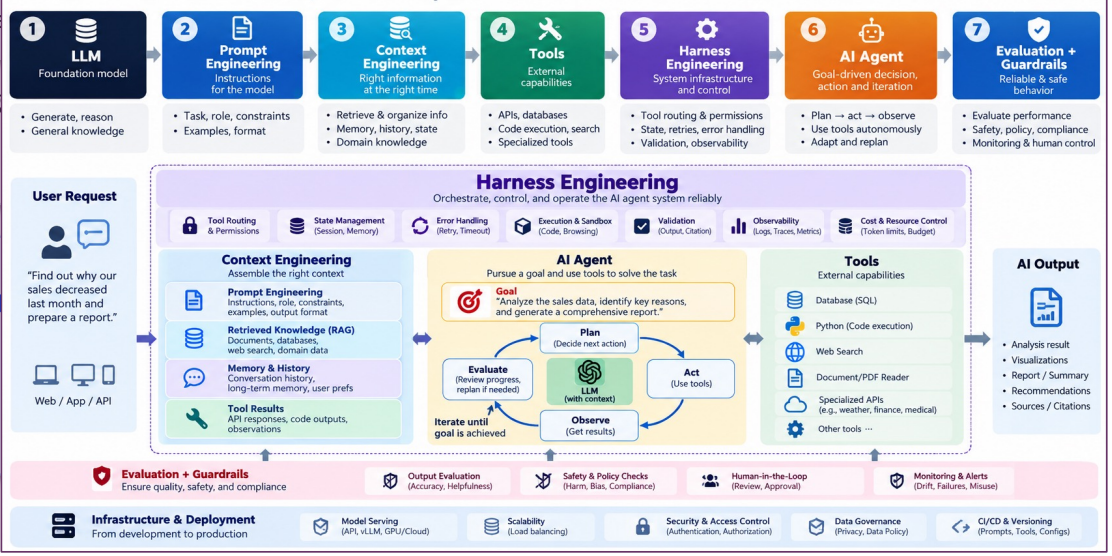
Evolution of LLMs

From language models to agentic models

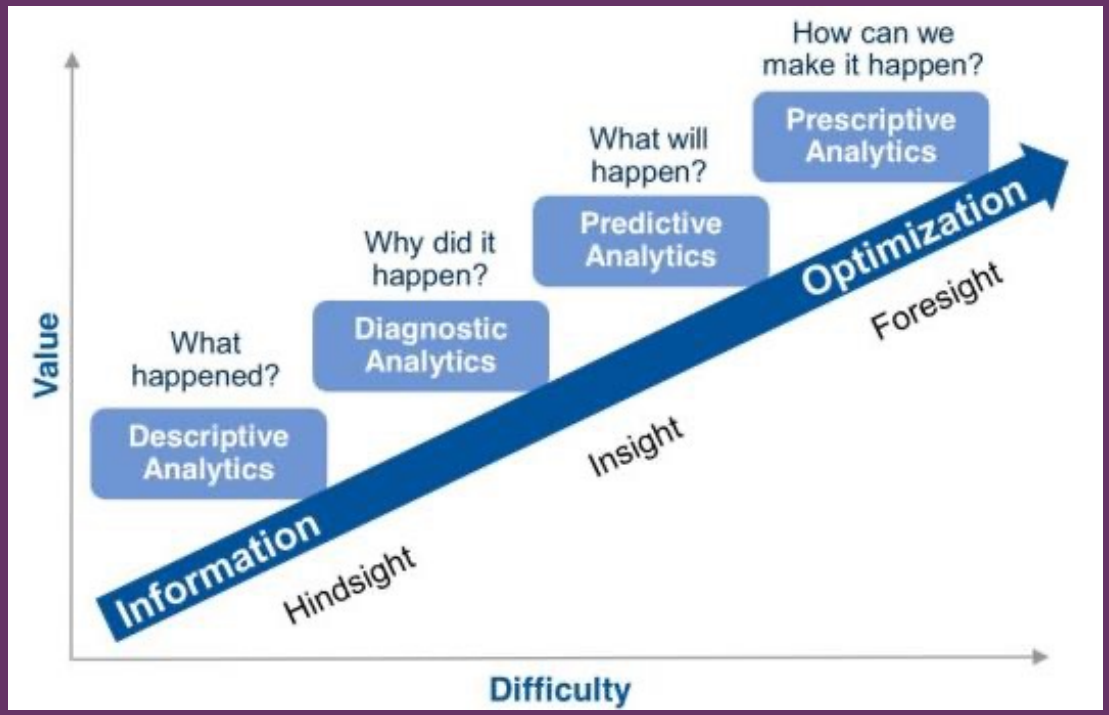
- **GPT-1/2** — Language Generation
- **GPT-3** — In-Context Learning
- Multimodality

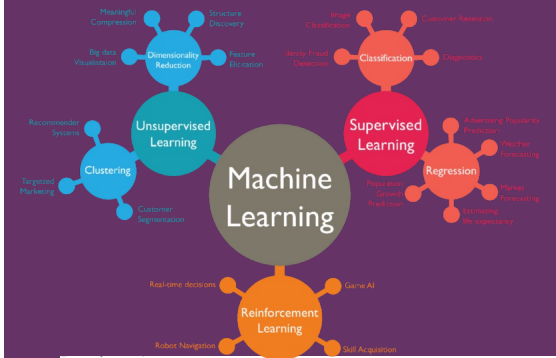
The Big Picture: Modern GenAI System

From a foundation model to a reliable, real-world AI system



Good luck! ☺


How can we make it happen?
What will happen?
Why did it happen?
What happened?
Prescriptive Analytics
Predictive Analytics
Diagnostic Analytics
Descriptive Analytics
Optimization
Foresight
Insight
Hindsight
Information
Value
Difficulty


Machine Learning
Unsupervised Learning
Supervised Learning
Reinforcement Learning
Dimensionality Reduction
Clustering
Classification
Regression


Sales Performance
$725,458
$501,240
$391,722
$678,781